

Types of Indexes in DBMS

Exam Reference Notes: Primary, Clustered, and Secondary Indexing

1. FUNDAMENTAL PREREQUISITES

Before classifying index types, it is essential to clarify how data ordering and attribute keys are defined in standard database indexing theory.

Ordered vs. Unordered Data

Ordered: Data records are physically sorted on disk in ascending or descending sequence (e.g., 1, 2, 3, 4...).

Unordered: Data records are stored without any physical sorting (e.g., 7, 2, 9, 1...).

Key vs. Non-Key Attribute

Key: Ordering/search attribute contains strictly unique values (no duplicates, e.g., Roll_No).

Non-Key: Ordering/search attribute contains duplicate or repeating values (e.g., Department).

2. PRIMARY CLASSIFICATION MATRIX ★

The core taxonomy of DBMS index types depends entirely on two factors: whether the underlying data file is **Ordered** and whether the search attribute is a **Key**.

Data Arrangement	Attribute Type (Key?)	DBMS Index Type
Ordered	Key (Unique)	PRIMARY INDEX
Ordered	Non-Key (Duplicates)	CLUSTERED INDEX
Unordered	Key (Unique)	SECONDARY INDEX
Unordered	Non-Key (Duplicates)	SECONDARY INDEX

3. DEEP DIVE INTO INDEX TYPES

A. Primary Index

Created when the search key specifies the physical sequential order of the file, and the field is a **primary/candidate key (unique)**.

Ordered: YES | Key: YES (Roll_No) → 1, 2, 3, 4, 5, 6

B. Clustered Index

Created when data is physically sorted on a **non-key field**. Equal values are physically grouped ("clustered") together on adjacent disk blocks.

Ordered: YES | Key: NO (Dept) → CSE, CSE, CSE, ECE, ECE, ME

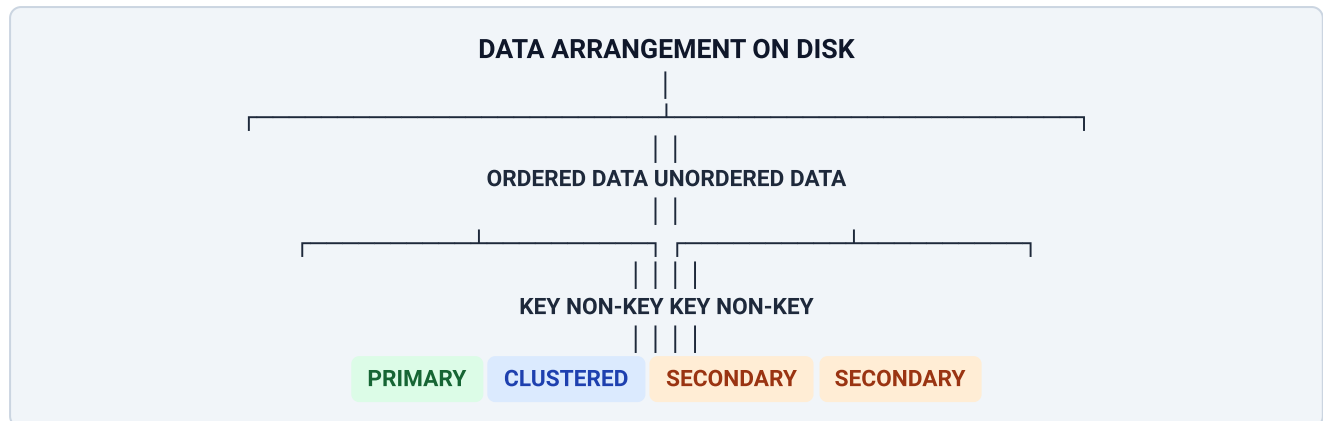
C. Secondary Index

Constructed over a field where the physical record sequence on disk does **not match** the search field order (Unordered data).

Unordered + Key (Roll_No): 7, 2, 9, 1, 5
Unordered + Non-Key (Dept): CSE, ECE, CSE, ME, ECE

4. 10-SECOND QUICK DECISION TREE

Use this flowchart logic to instantly solve classification questions in competitive exams:



5. MEMORY TRICKS & MNEMONICS

🧠 Memory Acronyms

- **P-K-O** → Primary = Key + Ordered
- **C-N-O** → Clustered = Non-Key + Ordered
- **S = Unordered** → Secondary handles all remaining Unordered cases!

6. COMPETITIVE EXAM TRAPS & FAQs

Q1. Are descending sorted order records considered "Ordered"?

Yes! Order refers to any strict deterministic sequence (ascending or descending). Both qualify for Primary/Clustered indexing.

Q2. Which index type handles multiple scenarios?

Secondary Index. It handles both *Unordered + Key* and *Unordered + Non-Key* cases.

Q3. What is the book index analogy?

Primary Index: Page numbers arranged sequentially (1, 2, 3...) at the top.

Secondary Index: Alphabetical subject index at the back (e.g., "DBMS → Page 50, SQL → Page 80").

7. GOLDEN RULES SUMMARY TABLE

Index Type	Data Sorted?	Unique Key?	Golden Rule
Primary	✓ Yes	✓ Yes	Ordered + Unique Key
Clustered	✓ Yes	✗ No	Ordered + Duplicate Values
Secondary	✗ No	✓ Yes / ✗ No	Any Unordered Data

